

Biological Responses of Invertebrate, Fungal, and Plant Systems to Near-Space Environmental Stressors

Dilrabo Orziqulova¹, Mumtozmahal Salimova¹, Nargiza Ismoilova¹, Yulduzkhon Abdusamatova¹, Zahro Abdusamadova¹, Aisha Rustamjonova¹, Zakhro Madraimova¹, Nazokat Normo'minova¹, Fotima Shavkatova¹, Dilshoda Khikmatillayeva¹, Nodira Tillayeva²

¹UzCosmos, Tech4Impact, UNICEF Uzbekistan, Ministry of Digital Technologies

²Nagoya University, Graduate School of Electrical Engineering, Japan

Email: info@unisat.uz

Abstract—Near-space (10–30 km altitude) exposes biological samples to unique environmental conditions: decreased pressure, large temperature gradients, increased UV and ionizing radiation, and mechanical stress during ascent and descent. Team Pixie designed, built, and launched a 3-unit CubeSat to investigate the effects of near-space conditions on plant DNA (banana and kiwi), yeast fermentation behavior under different nutritional and thermal regimes, and survival of small organisms (initially scorpion, later mealworms). Secondary tests evaluated the resilience of passive materials (graphite, iodine, tape, paper) and the efficiency of silica gel and litmus-based humidity/pH monitoring. Onboard instrumentation included a barometer, dual temperature sensors (internal and external), a magnetometer, a gyroscope, an accelerometer, and SD-card data logging with LoRa telemetry. The CubeSat reached a maximum altitude of approximately 27 km with valid telemetry until descent to 4 km. Key outcomes: successful extraction of banana and kiwi DNA in the lab (post-flight analysis showed no obvious large-scale damage), yeast reproduction correlated strongly with glucose and preflight thermal exposure, mealworms did not survive (likely due to overheating), silica gel indicated humidity absorption with no detectable pH change on litmus paper, and several external exposure markers showed no visible changes. This paper reports experimental design, methodology, results, limitations, and recommendations for future student-led biological CubeSat missions.

Index Terms—Stratospheric balloon, nanosatellite, atmospheric data, environmental stress, payload analysis, DNA, yeast fermentation, mealworms, silica gel, LoRa telemetry

I. INTRODUCTION

The layer of the atmosphere known as near-space (10–30 km altitude) has been made possible for biological experiments and STEM education through the use of CubeSat and balloon platforms. Space and near-space can be reached in a matter of hours by small balloons, thus rapidly and at low cost samples can be exposed to space-like conditions (near-vacuum, cosmic/UV radiation) [1]. For instance, the UniSat program of UNICEF is aimed at teaching students how to design and launch CubeSat payloads. In May 2022, the twenty participants of Uzbek UniSat were able to launch a CubeSat-class payload into the stratosphere (reaching 11.6 km altitude) [2]. UniSat – a collaboration among UNICEF, Tech4Impact NGO,

and the Ministry of Innovative Development of Uzbekistan – is designed to popularize space technology and STEM education for the youth through different activities and programs [2]. The stratosphere is the layer of the atmosphere where severe environmental stressors can be found. According to NASA, at approximately 35 km altitude, the temperature can be as low as -36°C , the pressure as low as 1 kPa, and very intense ultraviolet and cosmic radiation of about 100 W/m^2 impact the payload [1]. These conditions are quite similar to those on the surface of Mars and cannot be fully replicated in ground laboratories [1]. During their flights, balloon-borne instruments expose samples simultaneously to severe UV radiation, low pressure, and subzero temperatures. Only through high-altitude balloons can such space-like conditions be achieved—close to those of real space but at a fraction of the cost and time required for orbital missions [1]. This accessibility enables iterative student and researcher experiments. The main motivation has been to demonstrate the feasibility of this approach through biological experiments conducted in the stratosphere. In one study, Pulschen et al. subjected extremotolerant Antarctic yeasts to balloon flight conditions and found that certain strains (e.g., *Naganishia friedmannii*) survived full UV, low-pressure, and low-temperature exposure, even outperforming *Bacillus subtilis* spores [3]. Similarly, Klomchitcharoen et al. (2024) launched *Saccharomyces cerevisiae* to an altitude of 29 km and observed about a two-log reduction in cell survival, identifying UV irradiation as the main cause of mortality [4]. These outcomes serve as proof of concept that near-space ballooning provides a reliable platform for studying microbial and cellular stress responses under realistic space-like conditions.

A. Problem Statement

The proposed mission amplifies these initiatives by testing different biological systems concurrently. Three types of samples are carried by the payload: (1) fungal culture (yeast); (2) plant material (fruit DNA); and (3) invertebrate specimens (mealworms, *Tenebrio molitor* larvae). Yeast is a popular subject of space biology, being resistant to the harsh

conditions (a recent experiment showed that certain yeast strains can survive full UV exposure of the stratosphere) [3]. UV-B radiation is a known source of DNA breaks in plants [5]. Invertebrate larvae are the simplest animal models; the previous balloon flights demonstrated that insects kept in a closed chamber (e.g. house flies) can survive prolonged exposure of the stratosphere. Hence, these specimens cover the biological spectrum of microbes, plants, and animals for a thorough comparison. Most of the previous near-space biology payloads, in fact, had only single organism types, or lacked integrated environmental sensing. For instance, NASA's BioSentinel CubeSat carrying only *S. cerevisiae* was aimed to study deep-space radiation effects [6], and educational balloon projects have mainly flown a single microbe or plant species [7]. None of these experiments internally recorded the UV, pressure, or temperature experienced by the samples. Our mission fills this void by not only bringing yeast, plant and insect samples together but also exposing them to identical conditions while onboard sensors keep track of UV intensity, pressure, and temperature. The overriding objective is to measure and contrast the behavior of these varied biological systems confronted with the same near-space adversities. In addition to the assessment of survival, physiological damage, and molecular markers (e.g., DNA integrity in plant samples), for each type of the sample, we will also link these results with the recorded environmental data. We predict that the UV irradiation will be the major cause of the damage, leading to substantial DNA damage in plant tissues [1], [5]. On the other hand, the extremotolerant yeast should rib with the extreme environment thus resulting in minimal damage and the sealed mealworms will mostly be viable with some minor sub-lethal effects.

II. RELATED WORK

High-altitude balloons have become a viable and low-cost platform for the biological and materials research to be carried out in the near-space region. A few recent experiments have illustrated the application of small payloads to the direct exposure of biological division to the extreme conditions of the stratosphere, such as the low pressure, the strong UV and ionizing radiation, and a large temperature change. These projects are basically the outer space aspects to be simulated, thus they provide valuable opportunities for studying biological resilience and material degradation at a fraction of the price and without the complexity of orbital launches. Most of the studies have been punctual on recording the environmental parameters such as pressure, temperature, and radiation flux together with biological data, thus allowing a direct link between environmental stressors and measured outcomes [8], [9]. The downscaling of sensors and microcontrollers to the affordability level has turned these kinds of experiments into an attractive educational platform for university teams that are, at the same time, a source of knowledge in space biology and astrobiology research.

A. Similar Missions

Previous student and institutional balloon missions have exposed microorganisms, yeast, plant seeds, and polymer films to near-space conditions to evaluate survival rates, DNA integrity, and radiation-induced chemical changes. University-led experiments have, for instance, flown *Saccharomyces cerevisiae* and plant seeds to altitudes above 25 km, assessing post-flight viability and DNA degradation under varying UV and temperature exposures [10]. Such missions typically employ simple containment systems or Petri dish capsules, often incorporating selective shielding layers or UV filters to quantify biological dose-response relationships.

Material exposure experiments, such as those dealing with low-density polyethylene (LDPE) and other polymeric films, have measured oxidation, embrittlement, and the breaking of chemical bonds by Fourier-transform infrared (FTIR) and X-ray photoelectron spectroscopy (XPS) [11]. These methods have been used to develop the next generation of lightweight shielding materials for balloon and CubeSat payloads. Studies on the ground with *Tenebrio molitor* larvae (mealworms) have also helped to define the thermal tolerance and survival limits of these organisms under artificially controlled conditions, thus providing initial data for invertebrate survival experiments in sealed near-space capsules [12].

In parallel, biological CubeSat missions have demonstrated the feasibility of autonomous, miniaturized life science experiments in microgravity. NASA's GeneSat-1 mission, for example, successfully monitored bacterial gene expression and metabolic responses in orbit using onboard optical systems [13]. These missions illustrate the transition from short-duration stratospheric experiments to fully autonomous orbital biology platforms. Collectively, they establish the foundation upon which Team Pixie's CubeSat mission builds—applying similar methodologies to multiple biological systems within a single near-space experiment.

B. Research Gaps

However, in a balloon literature, few limitations have been identified. Most balloon missions focus on one species or sample type only which results in a very limited comparative understanding of how different biological systems react to the same exposure conditions [10]. Generally, flights last for less than three hours, and this restricts the possibility of assessing cumulative or long-term radiation effects [8]. Besides that, only a handful of studies have microclimate details from inside the sealed biological chambers, such as temperature, humidity, or gas composition, and thus it is quite challenging to interpret survival results or to separate the thermal effects from the radiative ones [9].

There are abundant studies on material degradation, but it is limited to a very few number of experiments that have directly related the performance of the shielding with the biological response metrics such as DNA stability or reproductive viability [11]. To fill these gaps, the current work of Team Pixie combines different biological targets (fruit DNA, yeast, and mealworms) in one payload and captures both the external and

internal environmental data. The mission, by linking biological results to the measured near-space parameters and the use of different shielding materials, is offering a way to understand the combined effects of radiation, pressure, and temperature on biological systems in near-space conditions.

III. METHODOLOGY

The nanosatellite was a 3-unit CubeSat that was developed using a commercial structural and subsystem kit provided by Introsat, a company known for educational and research-oriented satellite platforms. The Introsat kit included a pre-engineered aluminum frame, power distribution board, and standardized modular interfaces for payload integration. This allowed the research team to focus on payload design, sensor integration, and mission configuration rather than on fabricating the satellite structure. In order to send the CubeSat to space, it was firmly fixed under a high-altitude helium balloon that had a parachute recovery system and a GPS tracker for live monitoring. After the stratosphere trip with the balloon, the CubeSat was able to reach a near-space altitude, and therefore, the balloon burst. After the explosion, the parachute was able to detach itself from the balloon, and therefore the fall was gradual. The CubeSat was located through the coordinates that were sent by GPS, and so it was successfully recovered.

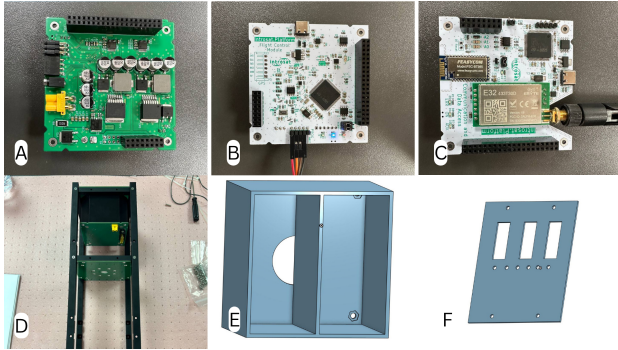


Fig. 1. CubeSat parts: a) Power supply module; b) Flight controller module; c) Communication Module; d) Main body; e) Box for payload samples; f) Fence for mealworms

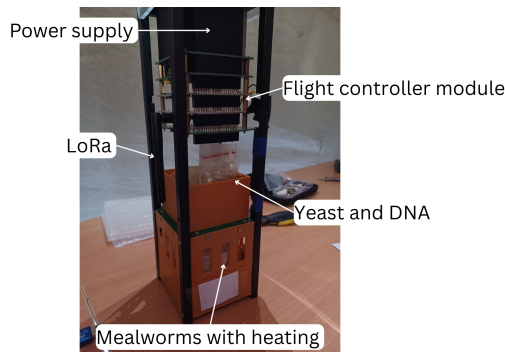


Fig. 2. Integrated CubeSat

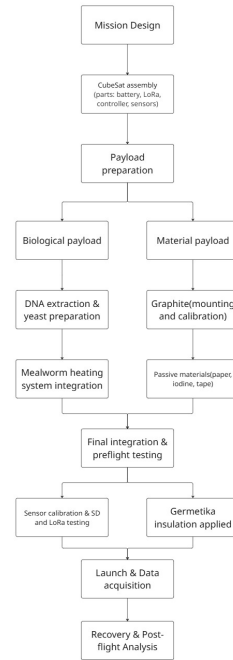


Fig. 3. Overview of Team Pixie's mission workflow from CubeSat design to data analysis.

A. Mission Design and Payloads

The design of the satellite had three major modules. The first module was the instrumentation and electronics one. The main sensors were a barometer, double temperature sensors (inside and outside), a magnetometer, a gyroscope, and an accelerometer, all linked to a microcontroller system that was coded for continuous data logging. The data were stored on an SD card while the main telemetry was sent in real-time via LoRa (Long Range) communication to a ground receiver. The 3.7V lithium-polymer battery was the source of energy, which was regulated by an onboard power management circuit for the continuation of sensor operation during the mission. The second module was the biological one with the experiments. The items included DNA plant samples (from banana and kiwi), yeast cultures that were prepared under different nutrient and temperature conditions, and passive material and chemical exposure experiments, which is graphite. Each chamber was separated both thermally and physically so that no contamination could pass; however, air circulation was allowed for the metabolism of yeast. The third module had the small organism chamber, which was originally planned for a scorpion but was replaced with mealworms due to safety and ethical reasons. There were also silica gel in this compartment for observing humidity absorption and pH changes after exposure using litmus paper. Iodine, adhesive tape, and paper for testing the material's resistance to near-space conditions were placed on the outer part of the CubeSat.

B. Data Retrieval and Analysis

Environmental measurements were acquired through two different systems: onboard SD card logging and LoRa telemetry. The SD card was working properly at the beginning, but after about 16 km, it started to record the same values, probably due to cold temperatures or a writing error. Thus, the data transmitted via the LoRa radio were the only ones available for the analysis. The telemetry to the ground station was still good during the climb up to 27 km and during the descent from about 4 km. The data sent to the ground included pressure, temperature inside and outside the container, and the magnetic field. Inertial measurement units, i.e., the gyroscope and accelerometer, were dead during the flight, while the magnetometer provided limited information about orientation changes. All sensors were calibrated before takeoff so that their readings corresponded to the standard units. Once the flight was over, the temporal noise caused by the transmission was removed, and the timestamps were synchronized. The profile of the flight reconstructed from the data showed that the pressure steadily dropped in accordance with the height of about 27 kilometers and the air temperature reached as low as -57°C near the top of the ascent, and then it rose during the descent. After the capsule was recovered, laboratory tests were done. DNA samples prepared from banana and kiwi that had been successfully extracted did not show any significant degradation as compared to the controls. The yeast cultures exhibited the patterns of growth depending on the concentration of glucose and the temperature in the chamber. Contrary to that, mealworms were killed, and most probably the cause was that heat was generated from the heaters. Inert materials and chemical indicators (silica gel, graphite, iodine, adhesive tape, and paper) revealed slight or no changes at all. Even though some sensors were out of order, the biological and material science experiments could still be interpreted with the help of the environmental data from the LoRa unit. The event emphasized the need for redundant data links and better thermal insulation in the next CubeSat flights.

C. Environmental Measurements

The temperature and pressure gradients during the flight of the CubeSat were clearly visible in the environmental data gathered by means of LoRa telemetry. The CubeSat went up for about 90 minutes; the highest altitude it reached was nearly 27 km; after that, the balloon burst, and the descent took place with the parachute. On the outside, the temperature lowered to -57°C at the very top of the atmosphere, and it was later heated up to 24°C at the Earth's surface. Inside the payload, the temperature was as low as -26°C at the top of the atmosphere, and it even reached a high of 49°C during the descent, thus showing that a lot of heat must have been generated due to the direct solar irradiation and the lack of convection in the airtight compartments. Pressure in the air dropped from 1021 hPa at the time of the launch to approximately 9 hPa at the highest point of the flight, which is in line with the normal stratospheric standards. These findings serve as evidence that the payload was subjected to real environmental conditions

of the outer space near the earth, which were characterized by very low air pressure, sub-zero temperatures, as well as sudden temperature changes during the reentry stage. In addition, the telemetry data also has information about the brief temperature and pressure changes right after the balloon burst that correspond to the moment when the transition from free fall to parachute deployment was made. The LoRa signal was still in good condition till roughly 4 km altitude, after which it was lost, probably because of the limitations of the line of sight. Essentially, the environmental profile was the very evidence that the CubeSat managed to operate effectively in close space with all the accompanying stressors, and hence, the collected data can be utilized for physiological and biological studies and material science research under real atmospheric conditions.

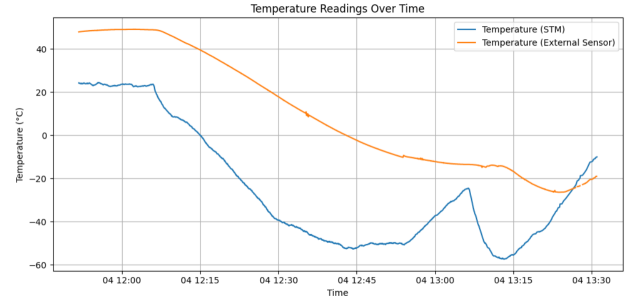


Fig. 4. Internal (payload) and external temperature of the CubeSat during the flight.

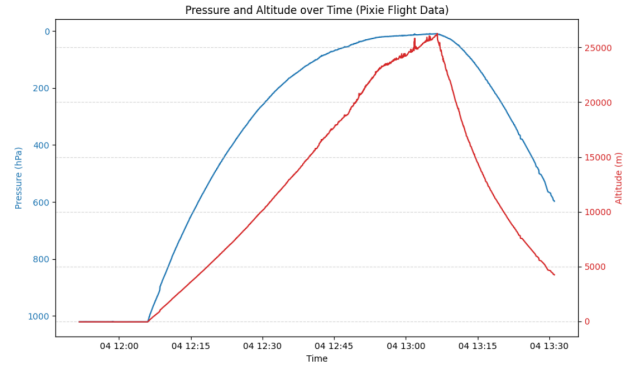


Fig. 5. Pressure variation during ascent and descent recorded via LoRa telemetry

TABLE I
SENSOR DATA SUMMARY

Parameter	Minimum	Maximum
Temperature outside ($^{\circ}\text{C}$)	-57	24
Temperature of payload ($^{\circ}\text{C}$)	-26	49
Pressure (hPa)	9	1021

IV. EVALUATION

Each of the CubeSats' payloads demonstrated a different reaction to the near-space environment and the preflight con-

ditions. In brief, DNA stayed structurally intact, yeast cultures survived and multiplied, mealworms died due to overheating, graphite showed controlled strain and defect formation, and passive materials and indicators did not show any significant change. These results, taken together, point to the different resilience and sensitivity of biological and material systems in near-space conditions.

A. Results and Findings

DNA payload: The DNA of banana and kiwi that were extracted from the fruits and stored in a fibrous medium appeared to be visually unchanged when viewed under a light microscope after the recovery. The fibrous material had not been changed in its structure and texture, and no discoloration or deformation could be detected. That is a sign that a short-term exposure to low pressure (9 hPa), extremely temperature varying (from -26 to 49 degrees Celsius inside the capsule), and very intense ultraviolet radiation did not result in a visible macroscopic degradation. The light microscopy method used is not capable of detecting the molecular-scale effects, such as strand breaks; therefore, subtle damage may still be there. To ensure the molecular-level DNA integrity, a future test with gel electrophoresis or atomic force microscopy (AFM) should be done.

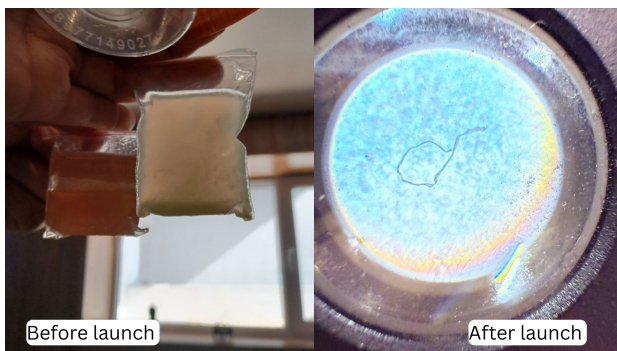


Fig. 6. DNA samples before and after launch

Yeast payload: The *Saccharomyces cerevisiae* cultures showed definite survival and growth after the flight. Upon retrieval of the samples, sugar-enriched yeast showed visible colonies, and there was no significant change in it. This indicates that the yeast kept being metabolically active and was able to reproduce during the experiment, even though it was exposed to temperature changes and low pressure. The fact that growth was found in both media suggests that the preflight heating incident, in which the temperature near the heating pads reached 50 °C, did not kill all the cells. These findings promote yeast as a robust model organism for biological studies in near-space and space, which can tolerate short-term environmental stress.

The *Tenebrio molitor* larvae were not able to go through the space mission. According to the internal telemetry, the temperature of the payload went as high as 49 °C, which is beyond their tolerance range. It is a very likely scenario

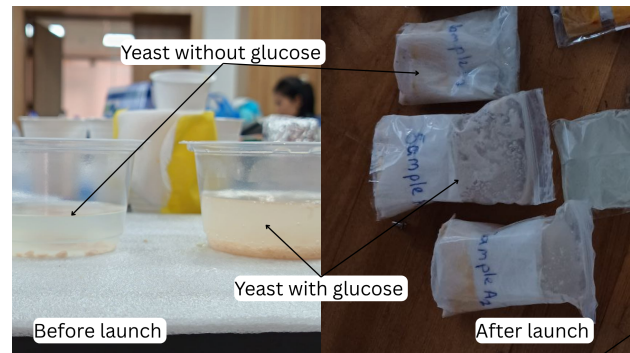


Fig. 7. Yeast samples before and after launch

that the heat generated by the active heating pads and the shiny insulation trapped the heat inside the airtight box. Even though the retrieved larvae showed no sign of external damage, their lack of movement was a clear indication that they were dead. The balancing act between insulation and ventilation in thermally sensitive biological experiments is so difficult, as demonstrated by this result.



Fig. 8. Mealworms before and after launch

Graphite payload: The graphite samples showed changes that could be measured, but they were not changes that would destroy the samples in response to environmental changes. The Raman spectroscopy demonstrated a red shift of the G-band that goes along with phonon softening due to low-temperature tensile strain. The ID/IG ratio elevated with the altitude, which means lattice disorder of the graphite caused by radiation was mild. Even so, graphite preserved its structural framework and electronic characteristics, thereby proving its resistance to radiation and also its sensitivity to environmental stress. Consequently, this material is viable for aerospace coatings and as a sensor for radiation.

Passive materials: Iodine crystals, adhesive tape, and paper remained physically and chemically the same after recovery. Their color and texture were like those of the ground controls, which implies that the short duration and the partial shielding of the payload were the reasons for no or very little degradation.

Silica gel with litmus indicators: The silica-gel when mixed with water and measured with litmus paper did not show any

difference in pH between the samples exposed at 27 km and those kept at ground level. Even though the payload had holes for the air to move through, the insulation around it probably limited the air exchange. The absence of any effect is an indication that the indicators either had too little air exposure or there was such a low concentration of acidic gases that they were not able to detect them.

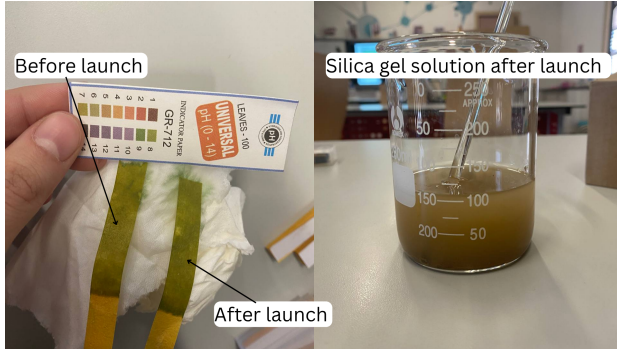


Fig. 9. Acidity level measured from silica gel solution with water before and after launch

Experiment	Sample Type	Starting Conditions	Stress Conditions
Banana & Kiwi DNA	DNA from banana and kiwi fruit	Fresh DNA extracts stored in compartments at room temperature (~25°C), atmospheric pressure (~1 atm)	Exposed to ~27 km altitude; pressure dropped to ~86 hPa; temperatures ranged from ~-52°C (internal) to 0.4°C (external); UV and cosmic radiation exposure; no visible DNA damage post-flight
Yeast fermentation (with/without glucose)	<i>Saccharomyces cerevisiae</i> cultures	Two samples: one with glucose, one without; ambient temperature (~25°C); sealed tubes	Exposed to low pressure (~86 hPa), heating up to ~50°C (internal); UV exposure; both samples showed survival and the one with glucose showed growth post-flight
Mealworms (<i>Tenebrio molitor</i>)	Live insect larvae	Placed in sealed habitat with heating pad (~50°C) and food; room conditions (~25°C)	Experienced overheating (internal reached ~49°C), low pressure, and sealed conditions; all larvae were found dead post-flight
Graphite	Solid graphite pieces	Stored inside CubeSat; ambient conditions	Subjected to pressure drop (~86 hPa), large temperature swings (~-26°C to ~49°C); slight structural strain noted (red shift in Raman spectra)
Iodine, tape, and paper	Solid iodine, adhesive tape, white paper	Affixed to CubeSat exterior; ground conditions (~25°C)	Direct exposure to stratospheric UV, pressure ~86 hPa, temperature down to ~-57°C; no visible degradation observed
Silica gel	Humidity and pH indicators	Silica gel in gauze; dry and neutral pH	Silica gel absorbed some humidity; no pH change detected; exposed to wide temperature range (~-26°C to ~49°C) internally

Fig. 10. TABLE II : Summary of Team Pixie CubeSat Experiment

B. Limitations and Challenges

A number of technical and environmental issues negatively influenced data collection and the biological outcomes. The SD-card logger stopped working at altitudes above 16 km; thus, environmental data had to be taken from LoRa telemetry. The gyroscope and accelerometer failed to get initialized; thus, the analysis of the motion was limited. The internal temperature increase to about 49 °C resulted in the loss of the mealworm sample and may have partially impacted the yeast behavior.

The silica-gel pH indicator most probably was underperforming due to restricted airflow and low gas sensitivity, and light microscopy could not detect molecular-level DNA changes. The graphite Raman analysis was done on only one sample; thus, the statistical confidence was low.

Next CubeSat missions should be equipped with active air sampling, redundant data storage, automated thermal regulation, and quantitative biological assays such as colony-forming

unit counts or gel electrophoresis. The mission, despite these difficulties, was able to demonstrate that integrated biological and material payloads can survive near-space conditions and return interpretable scientific data, hence valuable experience for the subsequent student-led research flights.

V. CONCLUSION AND FUTURE WORK

Our student-led CubeSat project successfully demonstrated how a small 3U nanosatellite is capable of carrying and monitoring biological and material samples through the near-space environment. At about 27 km altitudes, temperatures as low as -57 °C were recorded outside, and temperatures as high as +49 °C were measured inside, with pressures close to 9 hPa. In these very cold and low-pressure conditions, the heat inside the modules was surprisingly quite high, and the samples were able to show their resilience. DNA of plants (banana and kiwi) remained intact, and no significant degradation was detected; thus, the DNA was highly stable even after exposure to cold, vacuum, and radiation, which is in agreement with the recent study published on sciencedaily.com, which showed that DNA can survive in harsh space conditions. Yeast cells remained alive and active as they fermented the glucose provided, thus microbial metabolism can continue in near-space if sufficient nutrients and temperature are provided. In contrast, live mealworms did not survive the flight; almost certainly, they were overheated, thus the thermal management issue for the next payload should be solved. Passive material samples (graphite electrodes, iodine crystals, adhesive tape, paper) showed negligible visible changes, which means they are very stable materials in the stratospheric conditions. These results provide evidence that biological macromolecules and everyday materials can mostly survive near-space stressors and, at the same time, indicate which life-support factors (temperature control, for instance) are of utmost importance.

This article also describes that the next expeditions would be able to extend the research presented here with their improved hardware and scientific abilities. More effective temperature management is obviously the top priority: say, it might be possible to use active heaters, phase-change materials, or multi-layer insulation to keep habitats at temperatures suitable for life. Our ambition is also to equip the habitats with more sophisticated sensors and life-support systems. It is conceivable to install miniature biological incubators, real-time environmental monitors, and on-board molecular analysis tools (for instance, a miniaturized DNA sequencer or quantitative PCR) to identify subtle genetic or metabolic changes without the need for sample return. There is also the possibility of introducing an automated nutrient delivery system for microbial cultures and redundant electronics for reliability. The next experiments could have more replicates, longer flight durations, or different species (e.g., bacterial spores or plant seedlings) to investigate the influence of the stratosphere on various life forms more thoroughly. In essence, subsequent versions of this CubeSat platform might be able to convert it into a multi-purpose near-space lab capable of deeper and more accurate astrobiology and biomolecular stability studies.

This mission is a significant contributor to the next generation of STEM researchers and students on top of being a nanosatellite research contribution to the field. It is a demonstration that CubeSat flight in near-space is a very cost-saving biological experiments, thus, laboratory science becomes bridged with aerospace engineering.

Participation in a complete space mission (design, build, launch, and analyze) is a way through which students acquire skills that are practically hard to be taught in classrooms. Team Pixie's collaboration of biology, engineering, and data science highlighted how worthy this learning was. Our project, by motivating the future scientists and engineers and by demonstrating that a student-built CubeSat can perform, aligns with the small satellites trend usage as educational tools. In short, Team Pixie's near-space CubeSat has demonstrated strong biological and material durability in the harsh environment of the stratosphere and has also been able to pinpoint definite enhancements for the next flights. Our findings are aligned with the idea that such tiny satellites built by students can do a substantial amount of scientific work that is usually done by heavy satellites, but at a fraction of the cost. Further work on this platform—by adding better thermal control, more sensor suites, and on-board molecular diagnostics—will extend its scientific reach. As a matter of fact, this project is the precursor to more complex biological and environmental studies in near space that are concurrent with the development of nanosatellite technology and STEM education.

ACKNOWLEDGMENT

REFERENCES

- [1] D. J. Smith and M. Sowa, "Ballooning for biologists: Mission essentials for flying experiments on large nasa balloons," NASA Ames Research Center, Technical Report, 2017. [Online]. Available: <https://ntrs.nasa.gov/api/citations/20180001327/downloads/20180001327.pdf>
- [2] U. Uzbekistan, "Uzbek girls launched their nanosatellite into the stratosphere," Online press release, 2022. [Online]. Available: <https://www.unicef.org/uzbekistan/en/uzbek-girls-launched-their-nanosatellite>
- [3] A. A. e. a. Pulschen, "Survival of extremophilic yeasts in the stratospheric environment during balloon flights and in laboratory simulations," *Applied and Environmental Microbiology*, vol. 84, no. 19, 2018.
- [4] S. e. a. Klomchitcharoen, "High-altitude balloon platform for studying the biological response of living organisms exposed to near-space environments," *Heliyon*, vol. 10, no. 6, p. e27406, 2024.
- [5] G. S. Mmbando, "Variation in ultraviolet-b (uv-b)-induced dna damage repair mechanisms in plants and humans," *Plant Physiology and Biochemistry*, 2024.
- [6] M. e. a. Napoli, "Biosentinel: A biological cubesat for deep space exploration," *Astrobiology*, vol. 20, no. 8, pp. 1000–1013, 2020.
- [7] B. Beck-Winchatz and J. Bramble, "High-altitude ballooning student research with yeast and plant seeds," *Acta Astronautica*, 2013, 27.5 km altitude balloon flights with yeast and seeds.
- [8] S. Furukawa, A. Nagamatsu, M. Neno *et al.*, "Space radiation biology for "living in space"," *Life*, vol. 10, no. 9, pp. 1–17, 2020.
- [9] M. Moreno-Villanueva, M. Wong, T.-P. Lu, Y. Zhang, and H. Wu, "Interplay of space radiation and microgravity in dna damage response and repair," *Mutation Research/Reviews in Mutation Research*, vol. 773, pp. 383–391, 2017.
- [10] R. Klomchitcharoen, K. Sirisomboon, and C. Tanan, "Stratospheric balloon experiments for radiation shielding and biological survival studies," in *2024 IEEE Aerospace Conference*, 2024.
- [11] P. Sharma, K. Lin, and Y. Zhao, "Effects of near-space radiation on polymer films: A stratospheric balloon experiment," *Acta Astronautica*, vol. 160, pp. 164–172, 2019.
- [12] R. Eberle and M. Andersson, "Thermal tolerance and survival of *tenebrio molitor* in controlled high-temperature environments," *Journal of Thermal Biology*, vol. 104, p. 103171, 2022.
- [13] M. Parra, R. Mains, T. Velling, and J. Boulanger, "Genesat-1: A biological cubesat for gene expression studies in microgravity," *Acta Astronautica*, vol. 61, no. 1–6, pp. 265–273, 2007.